# Week 10
# Lecture 13
# Discovering Requirements

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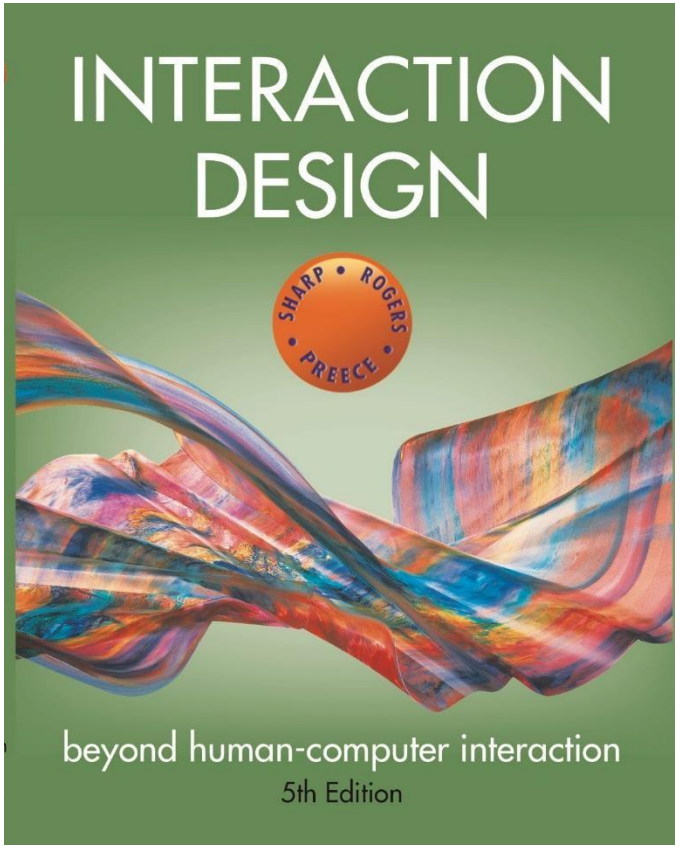
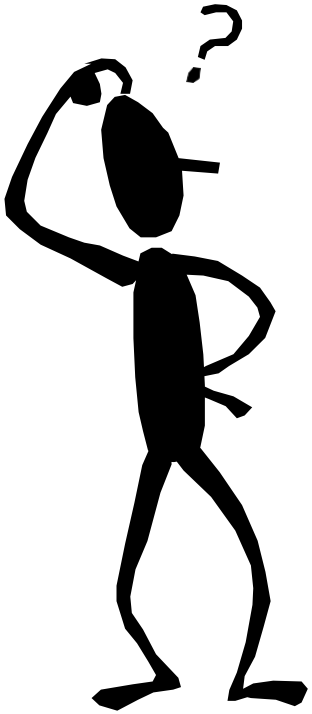

INTERACTION
DESIGN
SHARP • ROGERS • PREECE
beyond human-computer interaction
5th Edition

# Chapter 11

# Overview

- The importance of requirements
- Different types of requirements
- Data gathering for requirements
- Bringing requirements to life
- Personas
- Scenarios
- Capturing interaction with user cases

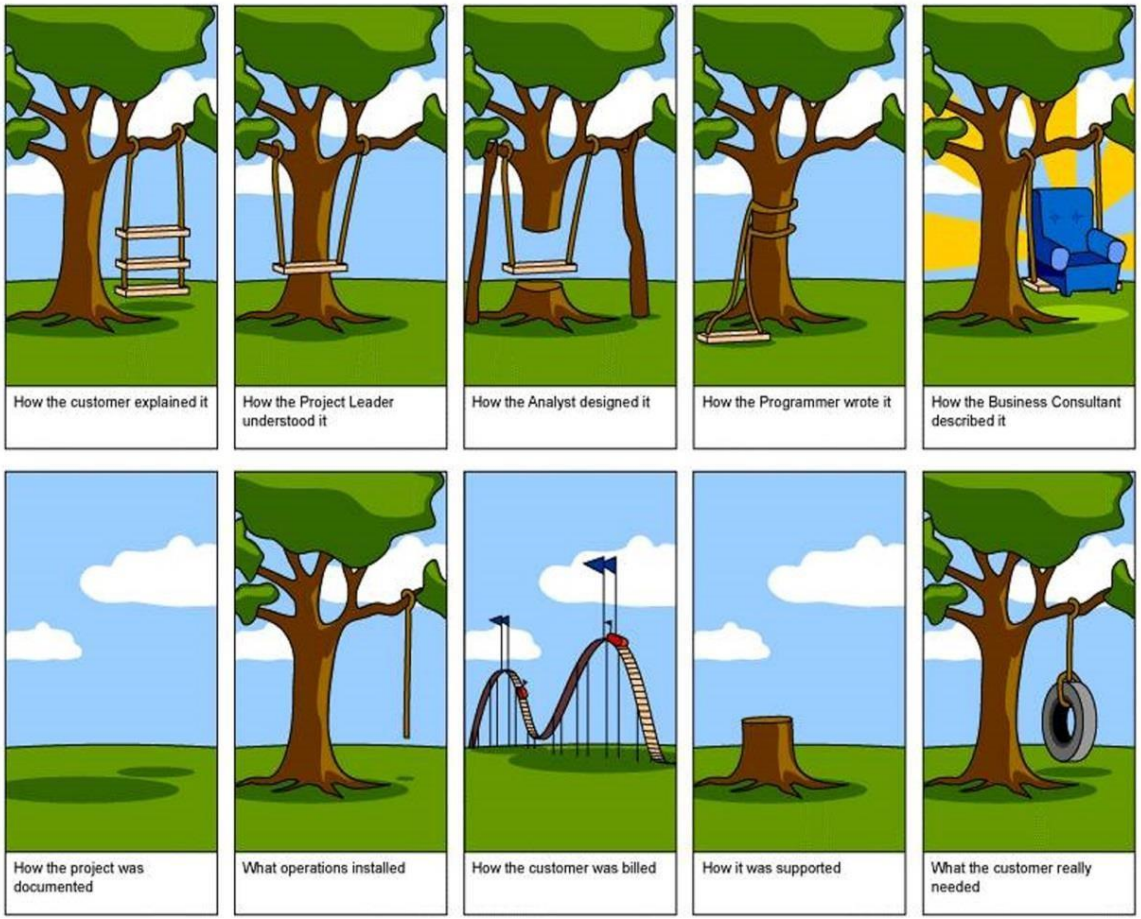


# What, how and why?

- What is the purpose of the requirements activity?
  - Explore the **problem space**
  - Establish a **description of what will be developed**
- How to capture requirements once discovered
  - In prototypes or operational product
  - Through structured or rigorous notations
  - In all cases, capturing requirements explicitly is beneficial in order to make sure that key requirements aren't lost through the iterations.

# Why bother?

Requirements activity is the stage where miscommunication occurs most commonly



# What are requirements?

- A statement about an intended product that specifies what it is expected to do or how it will perform
- Different forms and different levels of abstraction
- User stories (most prevalent in agile development contexts)
- Format:
  - As a <role>, I want <behavior> so that <benefit>
- Example user stories for a travel organizer might be:
  - As a <traveler>, I want <to save my favorite airline for all my flights> so that <I will be able to collect air miles>
  - As a <travel agent>, I want <my special discount rates to be displayed to me> so that <I can offer my clients competitive rates>

# Requirements: Volere shell – Req Example (1)

Requirement #: **75** Requirement Type: **9** Event/use case #: **6**

Description: **The product shall issue an alert if a weather station fails to transmit readings.**

Rationale: **Failure to transmit readings might indicate that the weather station is faulty and needs maintenance, and that the data used to predict freezing roads may be incomplete.**

Source: **Road Engineers**
Fit Criterion: **For each weather station the product shall communicate to the user when the recorded number of each type of reading per hour is not within the manufacturer's specified range of the expected number of readings per hour.**

Customer Satisfaction: **3** Customer Dissatisfaction: **5**

Dependencies: **None** Conflicts: **None**

Supporting Materials: **Specification of Rosa Weather Station**

History: **Raised by GBS, 28 July**

Volere

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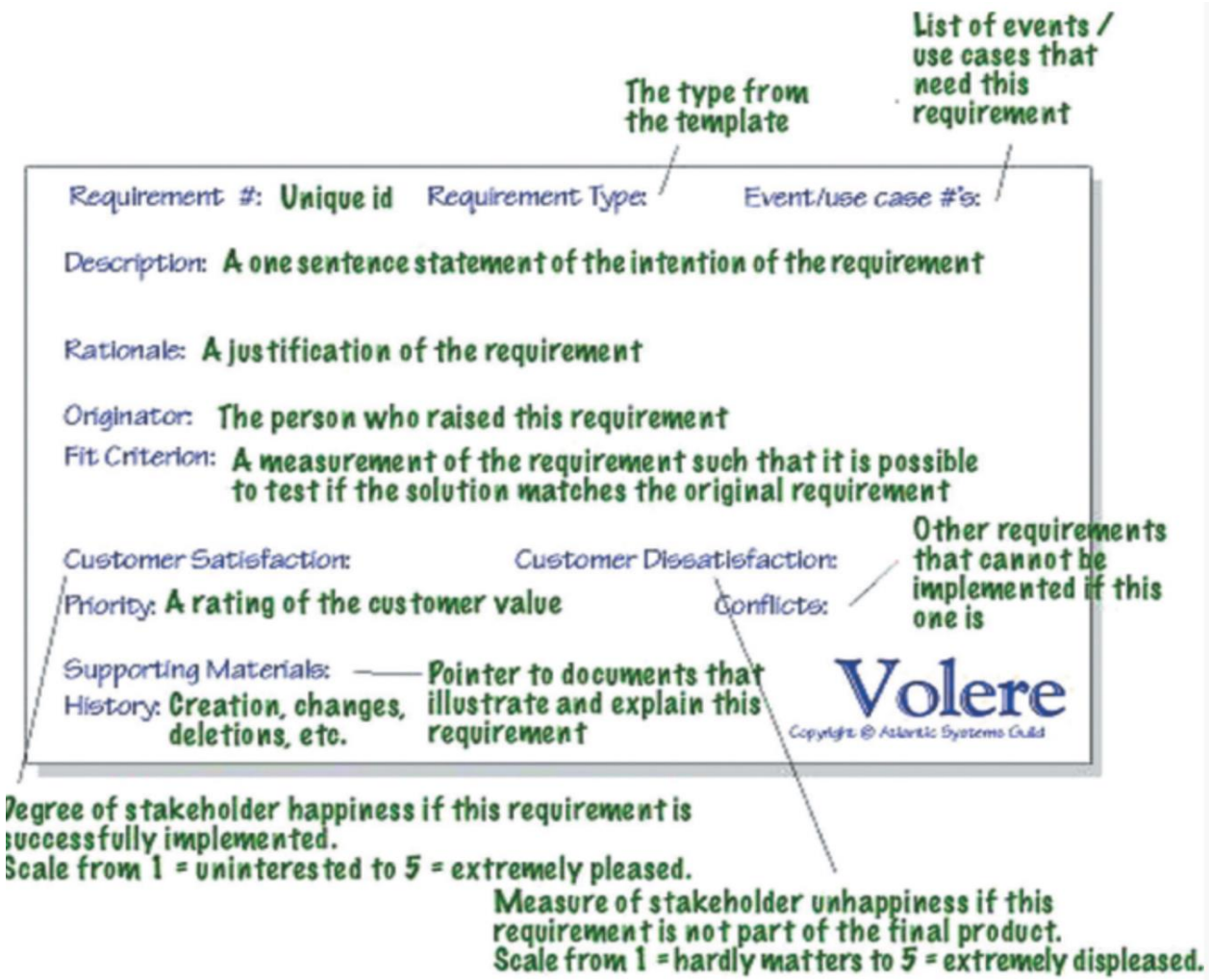
# Requirements: Volere shell – Req Example (2)



# Requirements: The seven product dimensions

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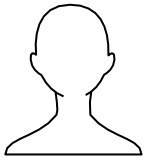
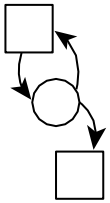
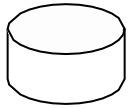
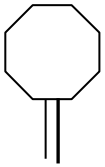
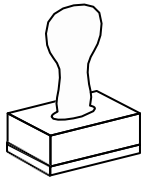
|  |  | |  |  | |  |
| User | Interface | Action | Data | Control | Environment | Quality Attribute |
| Users interact with the product | The product connects to users, systems, and devices | The product provides capabilities for users | The product includes a repository of data and useful information | The product enforces constraints | The product conforms to physical properties and technology platforms | The product has certain properties that qualify its operation and development |

*Source:* Gottesdiener and Gorman (2012), p.58. Used courtesy of Ellen Gottesdiener

# Requirements: Kinds/Types (1)

1. Functional:
    - What the system should do
2. Data:
    - **What** kinds of data need to be stored?
    - **How** will they be stored
    - For example, database

# Requirements: Kinds/Types (2)

## 3. Environment or context of use:

- **Physical**:
  dusty? noisy? vibration? light? heat? humidity? …. (for example, in a hospital)
- **Social**:
  collaboration and co-ordination, data sharing, distributed, synchronous or asynchronous, privacy
- **Organizational**:
  user support, communications structure and infrastructure, availability of training
- **Technical**:
  On what technologies will it run or need to be compatible?

# Requirements: Kinds/Types (3)

4. Users — Who are they?

- **Characteristics**:
  nationality, educational background, attitude to computers
- **System use**:
  - Novice: prompted, constrained, clear
  - Expert: flexibility, access/power
  - Frequent: shortcuts
  - Casual/infrequent: clear menu paths
- User profiling

# Requirements: Kinds/Types (4)

5 and 6. Usability and user experience goals

- Different products have different requirements and may be implemented in different ways, for example, trustworthiness

# Requirements: Data gathering (DG)

- Interviews, observation, and questionnaires
- It is usual for more than one data gathering techniques to be used in order to provide different perspectives.
- Studying documentation:
  - Procedures and rules are often written down in manuals
  - Good source of data about the steps involved in an activity and any **regulations** governing a task
  - Not to be used in isolation
- Researching similar products:
  - Good for prompting requirements

# Reqs: DG: Using probes to engage with users (1)

- Many types of probe:
  - Designed to prompt users into action
  - For researchers to learn about users
- Cultural probe:
  - Wallet containing postcards, maps, camera, photo album, and diary
  - Participants asked to answer questions using wallet contents
- Design probe:
  - Form relates specifically to particular question and context

Source:Wallace et al. (2013) Figure 6. Reproduced with permission of ACM Publications.

# Reqs: DG: Using probes to engage with users (2)

- Technology/Provocative probe:
  - Toolkits, mobile phone apps, sensor-based monitoring, for example, M-Kulinda to alert participants about unexpected movement at home.

# Reqs: DG: Contextual Inquiry (1)

- Part of Contextual Design, but also used on its own to gather requirements

- One-on-one field interviews (contextual interviews)
  - 1.5 to 2 hours long
  - Focus on daily life at home or work relevant to the project
  - Uses a model of master (participant) and apprentice (researcher)

# Reqs: DG: Contextual Inquiry (2)

- Four main principles:

  **Context:**

  Going to the user, wherever they are, and seeing what they do as they do it

  **Partnership:**

  User and interviewer explore user's life together

  **Interpretation:**

  Observations interpreted by user and interviewer together

  **Focus:**

  Project focus to understand to what should be paid attention

# Reqs: DG: Contextual Inquiry (3)

- Interview guided by "cool concepts" divided into two groups

- Joy of life concepts:
  - How products make our **lives richer and more fulfilling**
  - Connection/relationship with the user, sensation (pleasurable moments), accomplish (empowers user)

- Joy of use concepts:
  - Describe **impact** of using the product
  - Direct in action, the hassle factor, and the learning delta

# Reqs: DG: Contextual Inquiry (4)

- Following interview, interpretation session
  - Contextual design models are created or consolidated
  - Most relevant models are chosen by team

# Reqs: DG: Brainstorming for innovation

- Include participants from a wide range of disciplines, with a broad range of experience
- Don't ban silly stuff
- Use catalysts for further inspiration
- Keep records. Capture every idea, without censoring
- Sharpen the focus
- Use warm-up exercises and make the session fun

# Bringing requirements to life (1)

- Augmenting the basic requirements expressed as stories, in Volere template, or in other form

- Personas
  - Rich descriptions of typical users, not specific people

- Scenarios
  - An informal narrative story, simple, 'natural', personal, and not generalizable

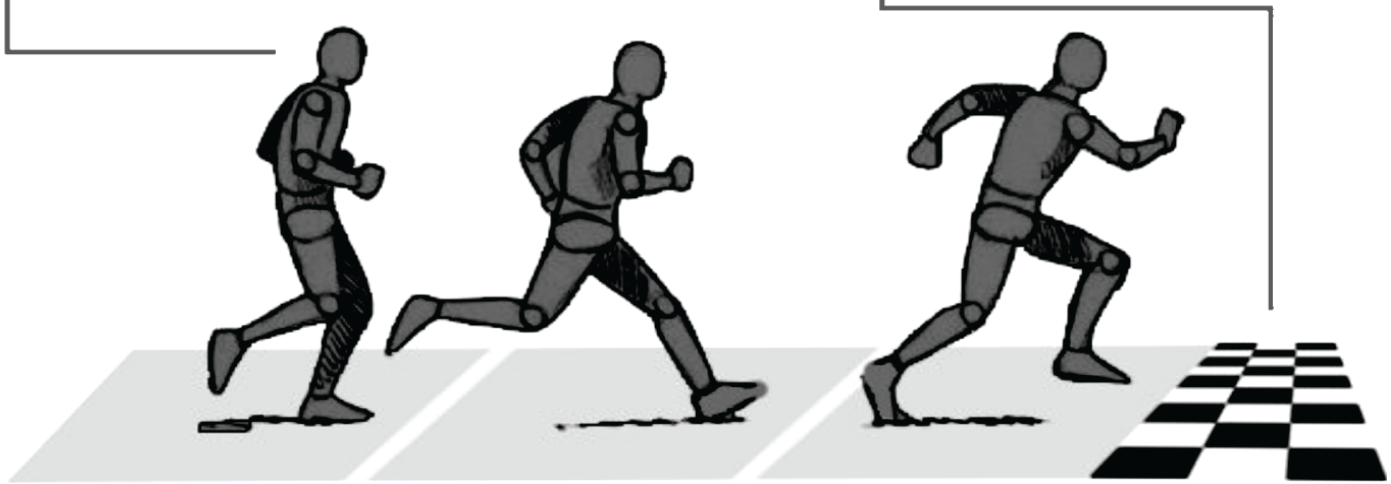
# Bringing requirements to life (2)

**1. Persona**

Defines who the story is about. This main character has attitudes, motivations, goals, and pain points, etc.

**3. Goal**

Defines what the persona wants or needs to fulfill. The goal is the motivation of why the persona is taking action. When that goal is reached, the scenario ends.



**2. Scenario**

Defines when, where, and how the story of the persona takes place. The scenario is the narrative that describes how the persona behaves as a sequence of events.

# Bringing requirements to life: Personas (1)

- Capture a set of **user characteristics** (user profile)
- Synthesized from **real people** based on user research
- Typical, not idealized

# Bringing requirements to life: Personas (2)

- Bring to life **with name**, characteristics, goals, and personal background
  - Relevant to product under development
- Good persona helps designer with design decisions and reminds team about who will use the product
- Develop a small set of personas with one primary

# Bringing requirements to life: Personas Example 1

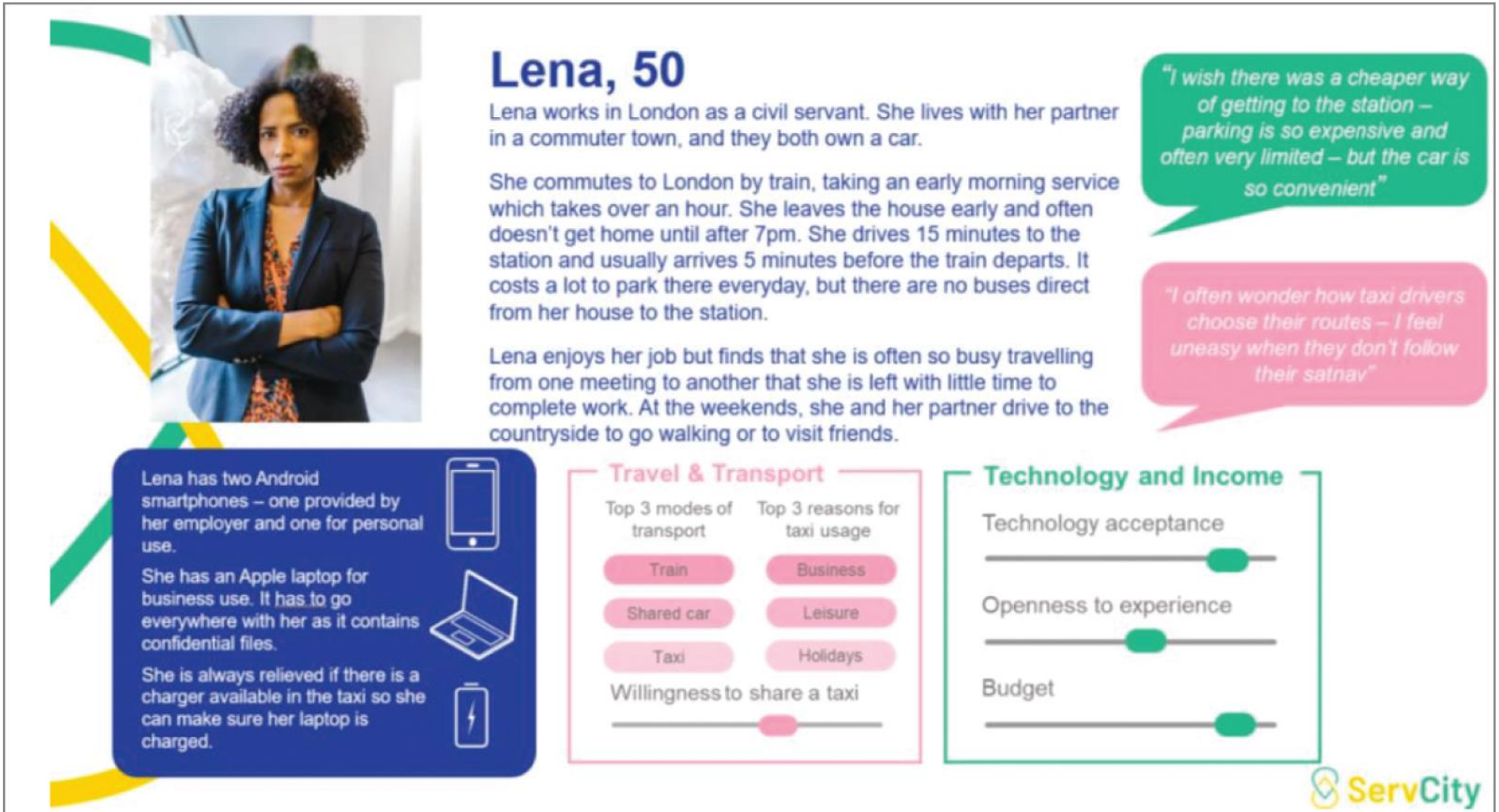



**Figure 5.** Example persona "Lena." Photo by RODNAE Productions from Pexels.

# Bringing requirements to life: Personas Example 2

## Family traveler

*""I want a travel organiser that will offer me a range of potential vacations that suit our needs"*

**Age:** 35
**Work:** Plumber
**Family:** Married, two children

### Personality

Introvert — Extrovert
Thinking — Feeling
Sensing — Intuition

Organised | Practical | Expects high standard

### Goals

- To book comprehensive travel quickly
- To find a trip that meets the needs of the whole family
- To feel supported and guided from the beginning of the booking experience right to the end.

### Frustrations

- Wasting time filling in forms
- Too much irrelevant information
- Existing systems tend to be too diverse and complicated

### Bio

Will loves to take his family on adventure holidays to explore new challenges. His children, Sky (8) and Eamonn (15) are old enough to take part in several sporting activities and he wants to make the most of this before they no longer want to go on trips with him and his wife, Claire. He likes the fact that choosing travel options is so much easier than it used to be, but is frustrated by the many different sources and disjointed options that this can result in. He wants a travel organiser that can provide clear support for family holidays while offering as wide a choice as possible.

### Motivation

Price
Comfort
Choice

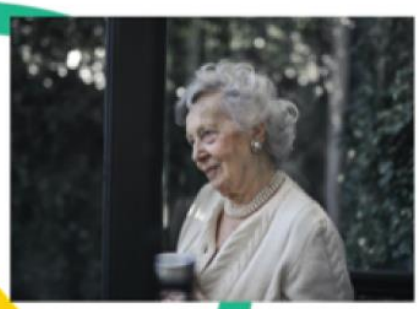
### Favourite destinations

Developed using Xtensio Templates

# Bringing requirements to life:Scenario for the navigation app of a large shopping center

Charlie wants to take his elderly mother, Freia, to his favorite home products store, ComfortAtHome. He knows that the store has moved within the shopping center, but he doesn't know where. He also needs to find a route that is suitable for his mother who uses a walker but doesn't like elevators. He opens the navigation app on his smartphone and enters the name of the store in the search feature. Two different branches of the store are listed, and Charlie asks for directions to the one nearest to their current location. A map of the shopping center is displayed, showing their current location, the location of the nearest store, and the suggested route. This route, however, includes a series of steps that are unsuitable for his mother. So, he asks for an alternative route that uses only ramps, which the app displays. They set off, following the new route provided.

# Bringing requirements to life: Scenario - Another Example (2)



## Mary
### Going to church

Mary likes to visit her family's church, once a month.

It takes two buses, but there is a 30 minute wait in between and she finds it hard to stand for long, especially in winter.

Friends have offered to take her, but she feels uncomfortable asking them to wait too long.

Instead she calls for a taxi, but makes sure to ask if the taxi driver can wait. It costs more, but at least she knows she can get home.

*"Il feel like they value my custom, even though I'm old!"*

Mary calls her local taxi company, she likes them because they always greet her by name and know where she wants to go. She's tried other companies, but found when they saw her address was 'sheltered housing', they often wouldn't show up.

*"I can relax now, knowing everything is confirmed"*

She likes to speak to the taxi operator directly, so she can arrange for the driver to wait for her at the church and then also bring her back after. She feels happier if she can confirm this with them and also reminds them not to send one of the bigger cars, which she struggles to step up into.

*"It's nice the driver offered to help, it's so awkward if you have to ask"*

After hearing 3 rings on her telephone she collects her coat, bag and stick and waits by the window. The taxi arrives and she is relieved to see the driver get out to help her with her things. She apologises for being 'so slow', but the driver is very relaxed and also helps her with her belt, which she always finds difficult to reach.

*"I'm glad the driver is patient and I don't feel rushed"*

The car is clean and smells fresh and the driver makes sure to speak up so she can hear him, which she appreciates. When they arrive at the church, the driver asks Mary the best spot to drop her so she doesn't have to walk too far. They agree for the driver to wait 30 minutes, which reassures her they won't leave without her!

**key journey stages**

- Planning
- Decision to book
- **Booking**
- **Waiting / Approach**
- **Ingress**
- **Transit**
- **Arrival**
- Payment
- Egress
- Feedback

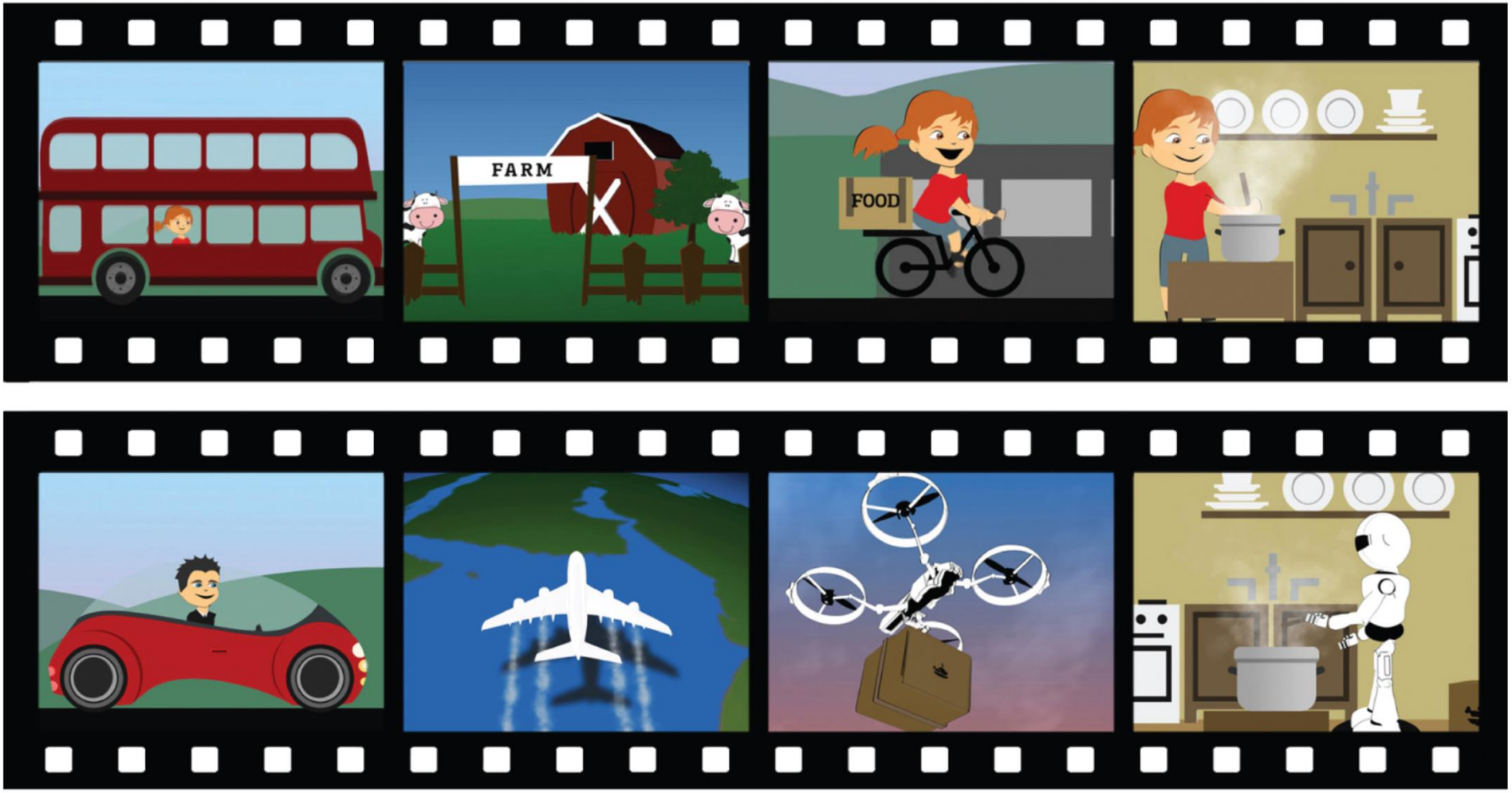
ServCity

**Figure 8.** Example scenario "Mary: Going to Church." Photo by Andrea Piacquadio from Pexels.

# Bringing requirements to life: Scenario Example (3)

| | |
|---|---|
| At home, Paula is notified by her smart home system that some of the groceries in her fridge are approaching their use-by date. She is encouraged to either consume them or donate them to a local food bank. She chooses the latter. | Paul does not have to care about food waste. Any food item that has gone out of date is automatically disposed of by his personal assistant. |
| Once back at home, Paula manually scans her newly acquired food items into her smart home system. This is to keep the system up to date on the available food and its use-by dates. | Every food item in Paul's fridge is being monitored by cameras and sensors. Whenever a particular food is running out, Paul is notified by an automated message and the food gets automatically restocked. |

# Bringing requirements to life: Scenario Example (4)



# Capturing Interactions with Use cases

- Focus on functional requirements and capture interaction
- Can be used in design or to capture requirements
- Use cases are step-by-step descriptions of interactions
- Two styles:
- Essential use cases: division of tasks, no implementation details
- Use case with normal and alternative courses: more detailed

# Example essential use case for travel organizer (1)

## *RetrieveVisa*

| USER INTENTION | SYSTEM RESPONSIBILITY |
|---|---|
| Find visa requirements | Request destination and nationality |
| Supply required information | Obtain appropriate visa info |
| Obtain copy of visa info | Offer info in different formats |
| Choose suitable format | Provide info in chosen format |

Note:
The user intention and system responsibility are offset vertically, showing a sequence of interactions

# Example essential use case for travel organizer (2)

**More detailed one**

1. The product asks for the name of the destination country
2. The user provides the country's name
3. The product checks that the country is valid
4. The product asks the user for their nationality
5. The user provides their nationality

# Example essential use case for travel organizer (3)

6. The product checks the visa requirements of that country for a passport holder of the user's nationality

7. The product provides the visa requirements

8. The product asks whether the user wants to share the visa requirements on social media

9. The user provides appropriate social media information

# Example essential use case for travel organizer (4)

## Some alternative courses:

- 4. If the country name is invalid:
  - 4.1:The product provides an error message
  - 4.2:The product returns to step 1
- 6. If the nationality is invalid:
  - 6.1:The product provides an error message
  - 6.2:The product returns to step 4
- 7. If no information about visa requirements is found:
  - 7.1:The product provides a suitable message
  - 7.2:The product returns to step 1

# Summary

- A requirement is a statement about an intended product that specifies what it is expected to do or how it will perform.
- Articulating requirements avoids miscommunication and supports technical developers and users to contribute.
- Different kinds of requirements: functional, data, environmental (context of use), user characteristics, usability goals, and user experience goals.
- Requirements data gathering uses: questionnaires, interviews, observation, studying documentation, and similar products
- Scenarios are a story-based narrative to explore existing behavior, potential of new products, and futuristic visions of use.
- Personas capture characteristics of typical users that are relevant to the product under development.
- Scenarios and personas together bring requirements to life.
- Use cases capture details about an existing or imagined interaction between users and the product.